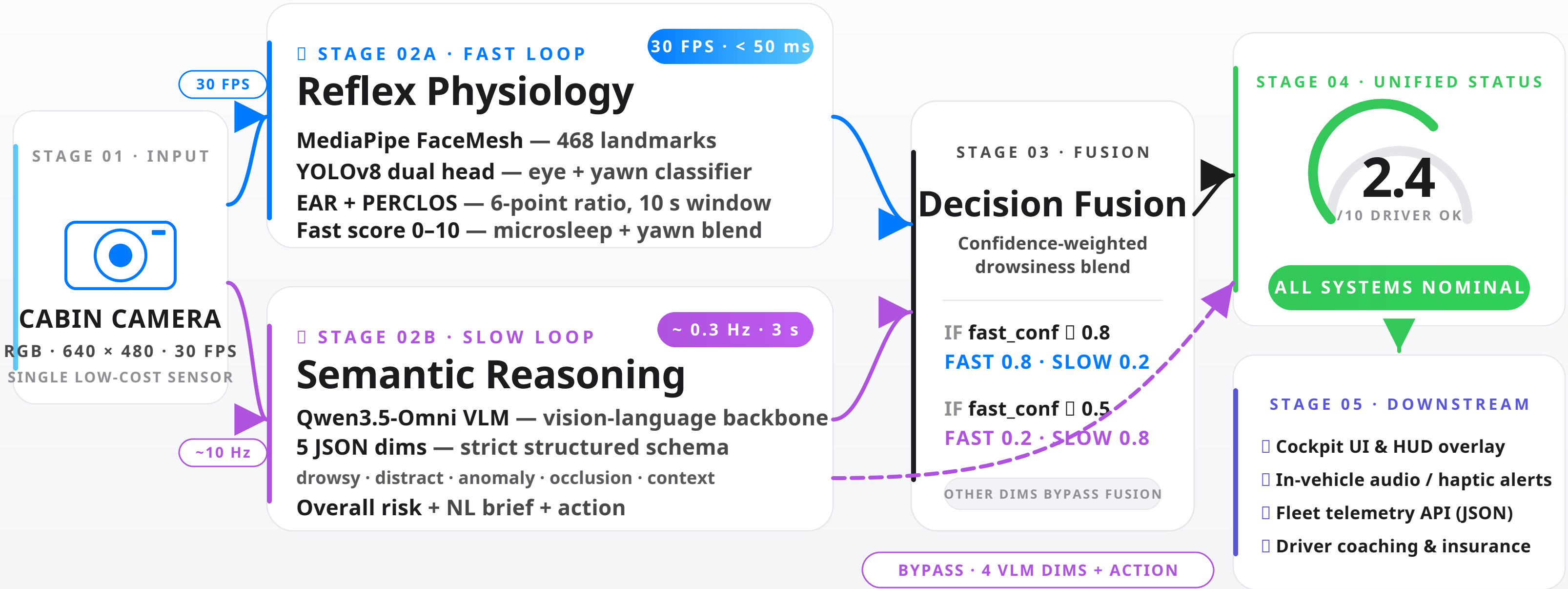


Dual-Loop Driver Monitoring Architecture

Millisecond-reflex physiology · paired with · continuous vision-language reasoning



COMPETITIVE DIFFERENTIATION

TRADITIONAL DMS

Eye closure only · no semantic context
High false-alarm rate

PURE-VLM PROTOTYPES

Rich semantics · but too slow (5–10 s)
Unsafe for safety-critical reactions

SENTINEL DMS

Fastest reflex + deepest reasoning
Unified by confidence-weighted fusion